

Operational Efficiency Transformation

HealthFirst Care Hospital

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Executive Summary

- Problem: 60+ minute wait times in Cardiology and Emergency depts.
- Solution: Digital transformation via automated scheduling & AI Triage.
- Key Metrics: 30% reduction in wait times, 15% increase in throughput.
- Strategy: Phased implementation with stakeholder-led change management.

Introduction

- Context: Multispecialty hospital with high patient volume.
- Opportunity: Overcoming manual bottlenecks that cause patient churn.
- Methodology: Analysis of 200 patient records and process benchmarking.
- Goal: Optimize resource allocation across 5 major departments.

Business Objectives

- Reduce wait times by 30% via automated resource flagging.
- Increase throughput by 15% without increasing staff headcount.
- Automate 90% of bookings through the Centralized Portal.
- Transition all triage to digital-first documentation.
- Maintain patient satisfaction scores at >8.5/10.
- Achieve \$50k annual saving in administrative paper costs.

Requirements: BRD & RTM

BRD Summary:

- Central Portal
- Shift Tracker
- EHR Sync

RTM Mapping:

- Goal 1 -> Feature: Auto-sms notification
- Goal 2 -> Feature: Load-balanced queuing

Stakeholder Analysis

- High Influence/High Interest: Hospital Board, Clinical Directors.
- High Interest/Low Power: Nursing Staff, Patients.
- Engagement: Weekly progress reviews with Board; Monthly training for Staff.
- Risk: Resistance to tech adoption mitigated by 'champion' training.

Scope Management Plan

- In Scope: Portal development, Staff training, Data migration.
- Out of Scope: Pharmacy inventory, Billing system replacement.
- Assumption: 99.9% network uptime during rollout.
- Constraint: \$500,000 fixed budget for Phase 1.

Process Mapping: As-Is vs To-Be

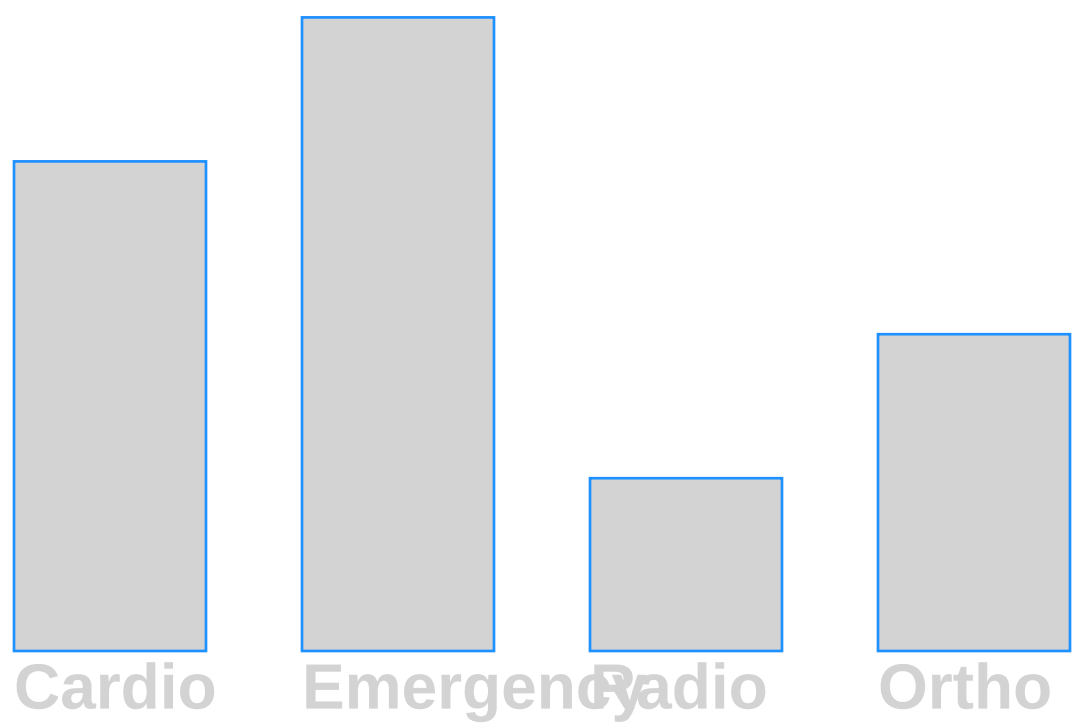
As-Is: Manual Call -> Paper Log -> Physical Queue -> Triage

To-Be: Online Portal -> AI Slotting -> Digital Queue -> Triage App



Data Analysis & Visualization

Key Data Trend: Peak congestion occurs between 10 AM and 12 PM



Risk Management

- Data Breach: Mitigation via End-to-End Encryption (AES-256).
- System Lag: Mitigation via Edge Computing and CDN.
- Lack of Adherence: Mitigation via incentivized training programs.
- Contingency: 24/7 technical support desk during first 3 months.

Key Findings & Recommendations

- Finding: Staffing levels ignore peak patient load times.
- Finding: Patients prefer digital booking (70% survey response).
- Recommendation: Deploy 'Dynamic Staffing' module by Q3.
- Recommendation: Consolidate Radiology scheduling with Ortho.

Conclusion

- HealthFirst Care is poised for a successful digital transition.
- The proposed plan aligns with business goals and budget constraints.
- Operational efficiency will likely improve by at least 25% in Year 1.
- Final Goal: Becoming a data-centric, patient-first facility.